

K654 (PRE-ARM) — The Confined-Field Casimir Is Flavor-Blind; It Cannot Supply the Down-Ladder’s Extra N_c^2 . Adjudication Criterion for the ζ -Landing.

Auditor: Keeper | Date: 2026-07-04 (ζ -computation pull) | Status: PRE-ARM — sets the fish-detection criterion before the forward ζ lands | Count 8

Pre-arm to adjudicate the forward ζ -computation. The physics likely splits the two receipts, and I need the criterion set *before* the result arrives.

The physics argument (why the ladder likely does NOT close via the Casimir)

- The down-ladder factor lives in m_s/m_d — a same-sector, FLAVOR/generation difference (s vs d).
- The confined-field Casimir (dressing) depends on COLOR + confinement radius, NOT flavor. s and d are both colored quarks with the same color charge $\rightarrow$ same color Casimir.
- A flavor-blind factor CANCELS in the same-sector ratio m_s/m_d . (This is exactly K650: the dressing cancels in same-sector ratios — *that’s why* $m_s/m_d = 20$ is 0σ pure-bare.)
- ☐ The confined-field Casimir contributes ~ 0 to m_s/m_d . It cannot be the source of the generation-dependent $2\cdot C_2$.

Likely verdict (my pre-arm, to be confirmed by the forward computation)

- RADIUS receipt: likely PASSES. $Z = C_F = 4/3$ is real — the confined-field Casimir *does* set the absolute dressing/size (an absolute quantity, not a same-sector ratio, so no cancellation). Forward-derivable, banks the charge-radius receipt.
- LADDER: likely does NOT close via the Casimir. The identity $2\cdot C_2 = N_c^2 \cdot C_F$ ($12 = 9 \times 4/3$) is likely numerical, not mechanistic — because the Casimir is flavor-blind and cancels in m_s/m_d , so the extra N_c^2 can’t come from it. And 12 is form-cheap ($2\cdot C_2$, $N_c^2 \cdot C_F$, $\text{rank}^2 \cdot N_c$, $\text{rank} \cdot C_2$... many readings). The bare-ratio $2\cdot C_2$ lives in the substrate deposit (K652), a different mechanism.

THE ADJUDICATION CRITERION (the fish-detector for the landing)

If the team reports “independent $N_c^2 \rightarrow$ ladder closes at 20,” I must verify the N_c^2 is FLAVOR/generation-dependent — i.e., it survives the same-sector ratio m_s/m_d . A color-Casimir N_c^2 is flavor-BLIND and cancels; a flavor-blind N_c^2 that nonetheless “closes” 20 is the double-count fish. The extra N_c^2 closes the ladder *only* if it distinguishes s from d — which the color Casimir, by construction, does not.

So the honest questions for the ζ -landing: 1. Does $Z = C_F$ forward (radius derived-strong)? — likely yes. 2. Is there a flavor-dependent N_c^2 in the confined-field structure that survives m_s/m_d ? — likely no (color Casimir is flavor-blind). If “yes” is claimed, demand the flavor-dependence explicitly, or it’s the double-count.

Net

I expect the ζ -computation to bank the radius and leave the down-ladder an honest negative — because the confined-field Casimir is flavor-blind and cancels in the same-sector ratio, so it cannot supply the generation-dependent $2\cdot C_2$. If the team surprises me with a genuinely flavor-dependent color multiplicity, I’ll bank it — but the criterion (survives m_s/m_d) is non-negotiable, and it’s how I catch a flavor-blind N_c^2 dressed as a forcing. Down-ladder’s real home for the $2\cdot C_2$ remains the substrate deposit (K651/K652), not the hadron bag.

— Keeper K654 pre-arm. Confined-field Casimir is flavor-blind $\rightarrow$ cancels in $m_s/m_d \rightarrow$ cannot supply the ladder’s N_c^2 . Likely: radius banks, ladder stays negative. Criterion: the closing N_c^2 must be flavor-dependent (survive the same-sector ratio) or it’s the double-count fish.